

Python for Data Science - 2305CS303

Lab - 9

Roll No. : 128

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1. Create a NumPy array of 10 student marks and display its dtype, size, ndim, shape, itemsize, and nbytes.

```
In [1]: import numpy as np
arr1= np.array([2,3,4,5,6,7])
print(arr1)
print(arr1.dtype)
print(arr1.size)
print(arr1.ndim)
print(arr1.shape)
print(arr1.itemsize)
```

```
[2 3 4 5 6 7]
int64
6
1
(6,)
8
```

2. Create NumPy arrays using zeros, ones, full, arange, linspace, and identity functions.

```
In [3]: import numpy as np

arr1 = np.zeros((3,4))
arr2 = np.ones((5,3))
arr3 = np.full((4,4),7)
arr4 = np.arange(2,20,2)
arr5 = np.linspace(1,50,5)
arr6 = np.identity(5)

print(arr1)
print(arr2)
```

```
print(arr3)
print(arr4)
print(arr5)
print(arr6)
```

```
[[0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]]
[[1. 1. 1.]
 [1. 1. 1.]
 [1. 1. 1.]
 [1. 1. 1.]
 [1. 1. 1.]]
[[7 7 7 7]
 [7 7 7 7]
 [7 7 7 7]
 [7 7 7 7]]
[ 2  4  6  8 10 12 14 16 18]
[ 1.  13.25 25.5 37.75 50. ]
[[1. 0. 0. 0. 0.]
 [0. 1. 0. 0. 0.]
 [0. 0. 1. 0. 0.]
 [0. 0. 0. 1. 0.]
 [0. 0. 0. 0. 1.]]
```

3. Create an array of numbers from 1 to 25, reshape it into a 5×5 matrix, and display its transpose.

```
In [4]: arr = np.arange(1,26).reshape((5,5)).transpose()
print(arr)
```

```
[[ 1  6 11 16 21]
 [ 2  7 12 17 22]
 [ 3  8 13 18 23]
 [ 4  9 14 19 24]
 [ 5 10 15 20 25]]
```

4. Generate marks of 20 students randomly between 0 and 100 and find their highest, lowest, and average marks.

```
In [5]: marks = np.random.randint(0, 101, 20)
print("Marks of 20 student ")
print(marks)
print("Highest marks ", np.max(marks))
print("Lowest marks ", np.min(marks))
print("Average marks ", np.mean(marks))
```

```
Marks of 20 student
[52 93 34 32  9 59 14  9 37 50 94 92 23 62 67  4 38 64 69 20]
Highest marks  94
Lowest marks   4
Average marks  46.1
```

5. Generate random numbers using `rand()`, `randn()`, `uniform()`, and `normal()`. Use a seed to reproduce the results.

```
In [10]: np.random.randint(42)
print(np.random.rand(6))
print(np.random.randn(3))
print(np.random.uniform(1,9, 3))
print(np.random.normal(20, 3, 2))
```

```
[0.14670614 0.76247957 0.40998818 0.31516112 0.18642806 0.51338896]
[ 0.04842142  0.39373842 -0.29527677]
[5.25981648 5.63548163 2.34768751]
[19.39719865 17.73964268]
```

6. Given roll numbers 101–120, randomly select 5 students without replacement and demonstrate shuffling and permutation.

```
In [5]: np.random.seed(10)

roll = np.arange(101, 121)

print(np.random.choice(roll, 5, replace=False))
print(np.random.permutation(roll))

shuffle = roll.copy()
np.random.shuffle(shuffle)
print(shuffle)
```

```
[109 114 103 108 120]
[111 112 119 101 103 115 104 107 120 113 102 108 109 116 118 106 117 105
 114 110]
[107 119 114 105 117 106 113 104 108 112 118 120 109 111 102 101 116 115
 103 110]
```

7. Given marks of 15 students, calculate sum, mean, median, minimum, maximum, variance, and standard deviation.

```
In [11]: marks = np.array([44, 65, 29, 92, 76, 74, 61, 69, 35, 60, 80, 72, 81, 45, 50])
print("Marks:", marks)
print("Sum:", np.sum(marks))
print("Mean:", np.mean(marks))
print("Median:", np.median(marks))
print("Minimum:", np.min(marks))
print("Maximum:", np.max(marks))
print("Variance:", np.var(marks))
print("Standard Deviation:", np.std(marks))
```

```
Marks: [44 65 29 92 76 74 61 69 35 60 80 72 81 45 50]
Sum: 933
Mean: 62.2
Median: 65.0
Minimum: 29
Maximum: 92
Variance: 310.82666666666665
Standard Deviation: 17.630276987803303
```

8. Given daily sales for 10 days, calculate total sales, average sales, cumulative sales, and cumulative product.

```
In [12]: sales = np.array([100, 150, 200, 120, 180, 250, 300, 220, 170, 210])

print("Total Sales:", np.sum(sales))
print("Average Sales:", np.mean(sales))
print("Cumulative Sales:", np.cumsum(sales))
print("Cumulative Product:", np.cumprod(sales))
```

Total Sales: 1900
Average Sales: 190.0
Cumulative Sales: [100 250 450 570 750 1000 1300 1520 1690 1900]
Cumulative Product: [100 15000 30000
00 360000000 6480000000 1620000000000
486000000000000 106920000000000000 -2703440737095516160
4126511494937706496]

9. Create two NumPy arrays and perform addition, subtraction, multiplication, division, floor division, modulus, and exponentiation.

```
In [13]: a = np.array([10, 20, 30, 40, 50])
b = np.array([2, 4, 5, 8, 10])

print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)
print("Floor Division:", a // b)
print("Modulus:", a % b)
print("Exponentiation:", a ** b)
```

Addition: [12 24 35 48 60]
Subtraction: [8 16 25 32 40]
Multiplication: [20 80 150 320 500]
Division: [5. 5. 6. 5. 5.]
Floor Division: [5 5 6 5 5]
Modulus: [0 0 0 0 0]
Exponentiation: [100 16000 2430000 655360
0000000
9765625000000000]

10. Create an unsorted array of student marks and display it in ascending and descending order. Also find the sorted indices.

```
In [14]: marks = np.array([75, 42, 89, 63, 95, 58])

print("Ascending:", np.sort(marks))
print("Descending:", np.sort(marks)[::-1])
print("Sorted Indices:", np.argsort(marks))
```

Ascending: [42 58 63 75 89 95]
Descending: [95 89 75 63 58 42]
Sorted Indices: [1 5 3 0 2 4]

11. Given the following NumPy array:

Write a program to display:

- First 5 elements
- Last 4 elements
- Elements from index 2 to 6
- Every alternate element

```
In [15]: arr = np.array([10, 20, 30, 40, 50, 60, 70, 80, 90, 100])

print("First 5:", arr[:5])
print("Last 4:", arr[-4:])
print("Index 2 to 6:", arr[2:7])
print("Alternate:", arr[::2])
```

```
First 5: [10 20 30 40 50]
Last 4: [ 70  80  90 100]
Index 2 to 6: [30 40 50 60 70]
Alternate: [10 30 50 70 90]
```

12. Given the following NumPy array:

Write a program to display:

- The array in reverse order
- Every second element in reverse order
- Last 5 elements in reverse order

```
In [16]: arr = np.array([10, 20, 30, 40, 50, 60, 70, 80, 90, 100])

print("Reverse:", arr[::-1])
print("Every second reverse:", arr[::-2])
print("Last 5 reverse:", arr[-1:-6:-1])
```

```
Reverse: [100  90  80  70  60  50  40  30  20  10]
Every second reverse: [100  80  60  40  20]
Last 5 reverse: [100  90  80  70  60]
```

13. Given following NumPy Array:

Write a program to display:

- First two rows
- Last two rows
- Alternate rows
- Rows from index 1 to 3

```
In [17]: arr = np.array([[10, 20, 30],
                        [40, 50, 60],
                        [70, 80, 90],
                        [100, 110, 120],
                        [130, 140, 150]])

print("First two rows:")
print(arr[:2])
```

```

print("Last two rows:")
print(arr[-2:])

print("Alternate rows:")
print(arr[::2])

print("Rows from index 1 to 3:")
print(arr[1:4])

```

First two rows:

```

[[10 20 30]
 [40 50 60]]

```

Last two rows:

```

[[100 110 120]
 [130 140 150]]

```

Alternate rows:

```

[[ 10  20  30]
 [ 70  80  90]
 [130 140 150]]

```

Rows from index 1 to 3:

```

[[ 40  50  60]
 [ 70  80  90]
 [100 110 120]]

```

14. Given following NumPy Array:

Write a program to display:

- First two columns
- Last three columns
- First three rows and last two columns
- A 2×3 sub-array from the center

```

In [18]: arr = np.array([[10, 20, 30, 40, 50],
                        [60, 70, 80, 90, 100],
                        [110, 120, 130, 140, 150],
                        [160, 170, 180, 190, 200]])

print("First two columns:")
print(arr[:, :2])

print("Last three columns:")
print(arr[:, -3:])

print("First three rows and last two columns:")
print(arr[:3, -2:])

print("2x3 sub-array from center:")
print(arr[1:3, 1:4])

```

First two columns:

```
[[ 10  20]
 [ 60  70]
 [110 120]
 [160 170]]
```

Last three columns:

```
[[ 30  40  50]
 [ 80  90 100]
 [130 140 150]
 [180 190 200]]
```

First three rows and last two columns:

```
[[ 40  50]
 [ 90 100]
 [140 150]]
```

2x3 sub-array from center:

```
[[ 70  80  90]
 [120 130 140]]
```

15. Given following NumPy Array:

Write a program to display:

- The matrix with rows in reverse order
- The matrix with columns in reverse order
- The matrix with both rows and columns reversed
- Alternate rows and alternate columns

```
In [19]: arr = np.array([[10, 20, 30, 40],
                        [50, 60, 70, 80],
                        [90, 100, 110, 120],
                        [130, 140, 150, 160]])

print("Rows reversed:")
print(arr[::-1])

print("Columns reversed:")
print(arr[:, ::-1])

print("Rows and columns reversed:")
print(arr[::-1, ::-1])

print("Alternate rows and columns:")
print(arr[::2, ::2])
```

```

Rows reversed:
[[130 140 150 160]
 [ 90 100 110 120]
 [ 50  60  70  80]
 [ 10  20  30  40]]
Columns reversed:
[[ 40  30  20  10]
 [ 80  70  60  50]
 [120 110 100  90]
 [160 150 140 130]]
Rows and columns reversed:
[[160 150 140 130]
 [120 110 100  90]
 [ 80  70  60  50]
 [ 40  30  20  10]]
Alternate rows and columns:
[[ 10  30]
 [ 90 110]]

```

15. Given an array of marks, write a program to display students who scored more than 75 marks.

```

In [20]: marks = np.array([65, 80, 72, 90, 55, 85, 70, 95])

print(marks[marks > 75])

[80 90 85 95]

```

16. Given an array, write a program to display elements that are greater than 30 and less than 70.

```

In [21]: arr = np.array([10, 25, 35, 45, 55, 65, 75, 85])

print(arr[(arr > 30) & (arr < 70)])

[35 45 55 65]

```

17. Given an array, write a program to display elements that are divisible by 5 and greater than 25, and then display elements that are either less than 20 or greater than 45.

```

In [23]: arr = np.array([10, 15, 20, 25, 30, 35, 40, 45, 50, 55])

print("Divisible by 5 and > 25:")
print(arr[(arr % 5 == 0) & (arr > 25)])

print("Less than 20 or greater than 45:")
print(arr[(arr < 20) | (arr > 45)])

Divisible by 5 and > 25:
[30 35 40 45 50 55]
Less than 20 or greater than 45:
[10 15 50 55]

```

In []: